

UPI FRAUD DETECTION

1. ABSTRACT

Digital payment systems such as UPI have become an important part of everyday financial transactions. The large volume of digital transactions also creates opportunities for fraudulent activities. Fraudulent transactions may be associated with unusual transaction amounts, abnormal transaction frequency, device changes, suspicious account behavior, or other irregular patterns.

This project focuses on analyzing UPI transaction data using statistical and exploratory data analysis techniques to identify possible fraud indicators. The dataset initially contains **51,000 transaction records and 12 features**. After removing **881 duplicate records**, the analysis is performed on **50,119 transactions**. Among these transactions, **2,467 are fraudulent**, representing approximately **4.92%** of the data analyzed.

The project uses descriptive statistics such as mean, media, standard deviation, quartiles and percentiles. Outlier detection is performed using **Box Plot, IQR method and Z-score analysis**. Pattern analysis is carried out for transaction amount, time of transaction, device usage, account age and transaction frequency. Hypothesis testing is also performed using **T-test, Chi-square test and ANOVA**.

The analysis identifies high-value transactions, abnormal activity from new accounts, and unusual device behavior as useful fraud-monitoring signals. The project concludes that fraud detection should not depend on a single variable; instead, multiple behavioral signals should be combined to improve fraud monitoring.

2. INTRODUCTION

UPI has made digital payments fast, convenient and accessible. Users can transfer money instantly using mobile devices and payment applications. However, the increasing number of digital transactions also increases the possibility of fraudulent activities.

Fraud detection is challenging because fraud represents only a small portion of total transactions. In the dataset analyzed, only **4.92% of transactions are fraudulent**, while the remaining transactions are non-fraudulent.

Fraudulent behavior may not always appear as an obvious transaction. A suspicious transaction may involve an unusually large amount, frequent transactions within a short period, multiple devices, or abnormal behavior from a newly created account.

Therefore, statistical analysis can help identify unusual transaction patterns and provide early warning signals to fraud investigation teams. This project analyzes UP

transaction data to understand normal and suspicious behavior and to identify statistical indicators that can support fraud detection.

3. PROBLEM STATEMENT

Digital payment platforms process a large number of transactions every day. Fraud teams cannot manually investigate every transaction.

The major problem is identifying transactions or users whose behavior differs significantly from normal transaction patterns.

The project therefore aims to answer the following questions:

- How many transactions are present in the dataset?
- What percentage of transactions are suspicious?
- What is the average transaction amount?
- What is the average transaction frequency?
- Are transaction distributions highly skewed?
- Which users appear statistically unusual?
- Are frauds concentrated at specific times?
- Do suspicious users change devices frequently?
- Are new users riskier?
- Which variables act as strong fraud indicators?
- Are high-value transactions always risky?
- Which customer segments require monitoring?

4. OBJECTIVES

The main objectives of the project are:

1. To understand and clean the UPI transaction dataset.
2. To identify missing values and duplicate records.
3. To calculate descriptive statistics for transaction behavior.
4. To identify unusual transaction amounts using outlier detection techniques.
5. To analyze transaction frequency and account age.
6. To investigate the relationship between transaction behavior and fraud.

7. To analyse time, device, amount, and frequency patterns.
8. To perform statistical hypothesis testing.
9. To identify important statistical fraud signals.
10. To provide practical recommendations for fraud monitoring.

5. DATASET DESCRIPTION

The dataset contains **51,000 initial transaction records** and **12 columns**.

Dataset Features

Feature	Description
Transaction_ID	Unique transaction identifier
User_ID	Identifier of the user
Transaction Amount	Amount involved in the transaction
Transaction Type	Type of transaction
Time_of_Transaction	Time at which transaction occurred
Device Used	Device used for the transaction
Location	Transaction location
Previous_Fraudulent_Transactions	Previous fraud history
Account_Age	Age of the user account
Number_of_Transactions_Last_24H	Number of transactions during the previous 24 hours
Payment_Method	Method used for payment
Fraudulent	Target variable indicating fraud

The **Fraudulent** column is used as the target variable.

6. DATA UNDERSTANDING AND PREPROCESSING

The dataset was first loaded and inspected to understand its structure.

Initial Dataset

- Total records: **51,000**
- Number of columns: **12**
- Duplicate records: **881**
- Records after duplicate removal: **50,119**

Missing Values

Missing values were found in several columns:

Column	Missing Values
Transaction_Amount	2,520
Time_of_Transaction	2,552
Device_Used	2,473
Location	2,547
Payment_Method	2,469
Fraudulent	0

The target variable contains no missing values.

Duplicate records were removed before performing the main analysis so that repeated transactions would not unnecessarily affect the statistical results.

7. FRAUD DISTRIBUTION

preprocessing, the dataset contains:

After **Total analysed transactions:** 50,119.

- **Fraudulent transactions:** 2,467
- **Non-fraudulent transactions:** 47,652
- **Fraud percentage:** 4.92%

This shows that fraudulent transactions form a relatively small portion of the overall dataset.

Interpretation

The dataset is therefore imbalanced toward non-fraudulent transactions. This is important because a fraud detection system should focus on identifying the small number of suspicious transactions without generating too many false alarms.

8. DESCRIPTIVE STATISTICAL ANALYSIS

Descriptive statistics were calculated for transaction amount and transaction frequency.

Transaction Amount Statistics

Statistic	Value
Mean	\$2,996.25
Median	\$2,524.10
Standard Deviation	\$5,043.93
Q1	\$1,270.55
Q3	\$3,787.24
90th Percentile	\$4,535.53
95th Percentile	\$4,788.84
99th Percentile	\$49,997.80
Skewness	8.45

Average Transaction Frequency

The average value of transactions during the previous 24 hours is approximately:

7.50 transactions

Interpretation

The mean transaction amount is greater than the median. The skewness value of **8.45** indicates that the transaction amount distribution is highly right-skewed.

This means that most transactions have relatively moderate amounts, while a smaller number of transactions have extremely high values.

Therefore, median, quartiles and percentile values are useful for analysing transaction behaviour along with the mean.

9. OUTLIER DETECTION

Outlier detection was performed to identify unusually high transaction amounts and abnormal behaviour.

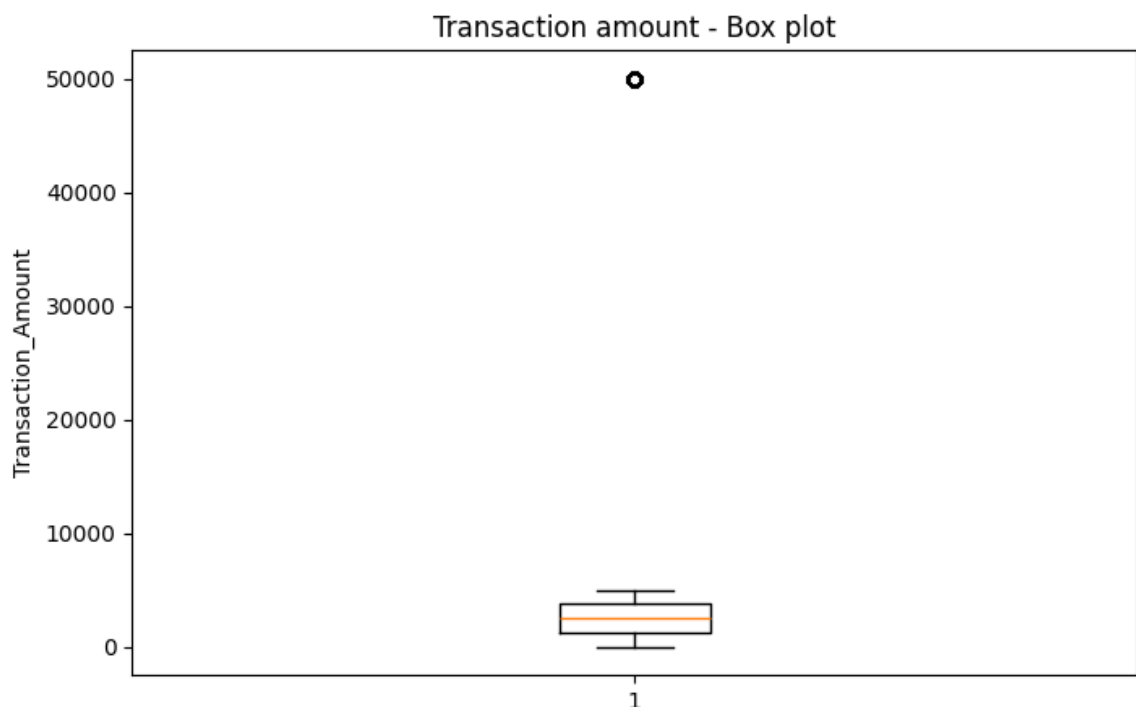
The following methods were used:

- Box Plot
- IQR Method
- Z-score
- Distribution analysis

9.1 Box Plot

The box plot was used to visually identify transaction amounts that are significantly different from the majority of transactions.

The plot shows that some transaction amounts are much higher than the normal range.



The box plot shows the distribution of transaction amounts and identifies unusually high transaction values. The extreme point near 50,000, indicates a very high-value transaction compared with the majority of transactions. This observation was further investigated using the IQR method, which identified 508 transaction-amount outliers.

9.2 IQR Method

The Interquartile Range method was used to detect extreme transaction amounts.

Calculated values:

- $Q1 = \$1,270.55$
- $Q3 = \$3,787.24$
- $IQR = \$2,516.69$
- Upper limit = $\$7,562.27$

Using the IQR method, **508 transaction records** were identified as transaction-amount outliers.

Interpretation

These transactions have unusually high amounts compared with the normal transaction distribution and can be treated as potential fraud-monitoring candidates.

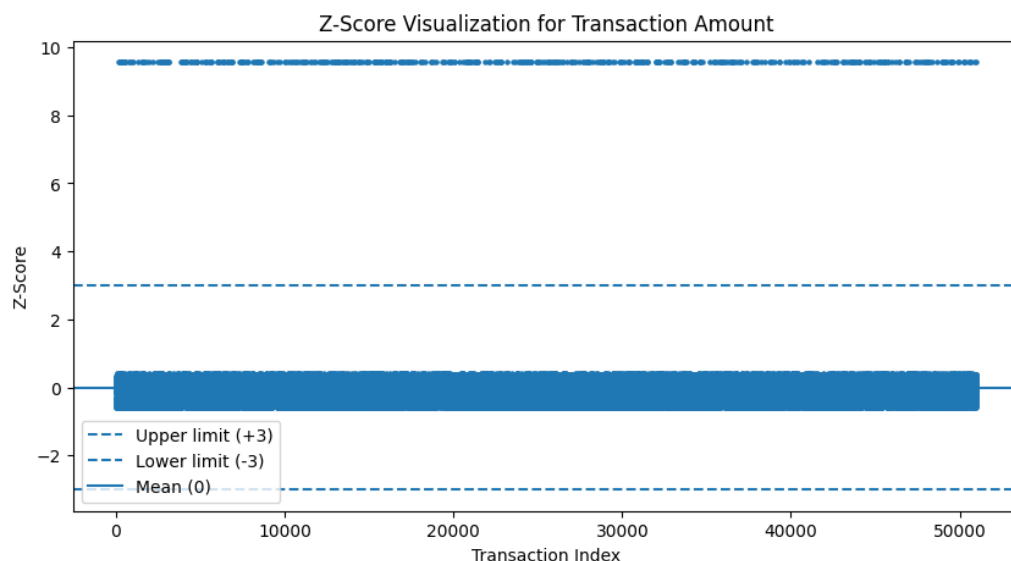
9.3 Z-SCORE METHOD

Z-score analysis was used to measure how far transaction amounts are from the mean in terms of standard deviations.

Extreme Z-score values indicate transactions that are statistically unusual.

A combined abnormality analysis also identified approximately **2,424 statistically unusual transactions**.

However, an unusual transaction does not automatically mean that it is fraudulent. It should be considered as a signal for additional investigation.



10. PATTERN ANALYSIS

The project investigates five major behavioural patterns:

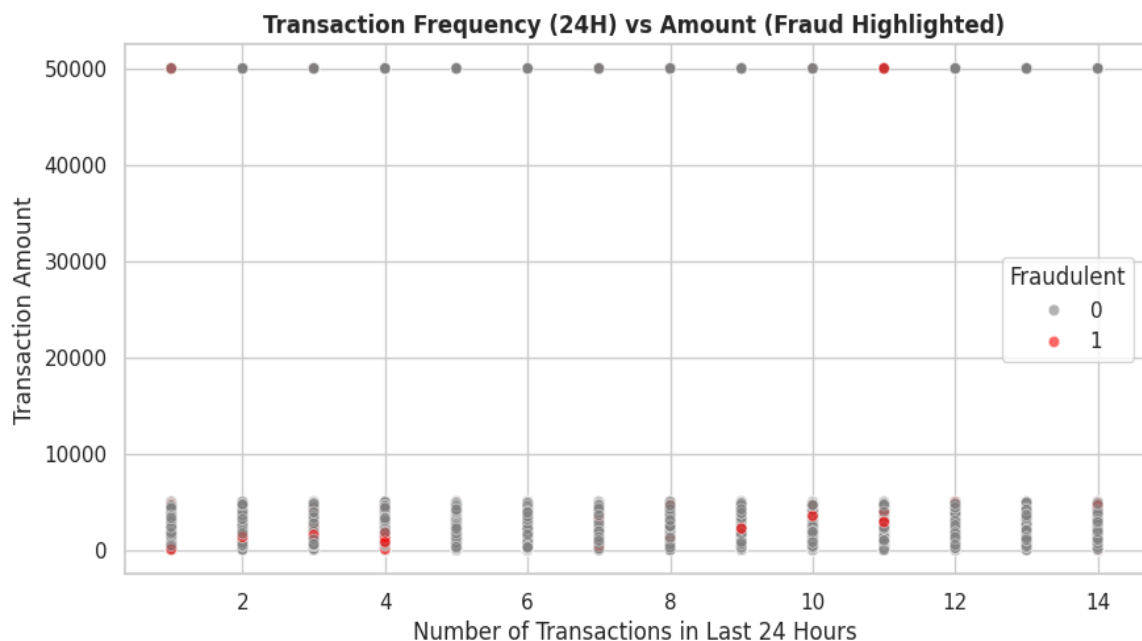
1. Transaction Amount vs Fraud
2. Time of Day vs Fraud
3. Device Changes vs Fraud
4. Account Age vs Fraud
5. Transaction Frequency vs Fraud

10.1 TRANSACTION AMOUNT VS FRAUD

Transaction amounts were analysed against the Fraudulent variable.

The analysis shows that extremely high transaction amounts are useful for identifying suspicious transactions.

However, the hypothesis test shows that transaction amount alone does not have a statistically significant difference between fraud and non-fraud groups.



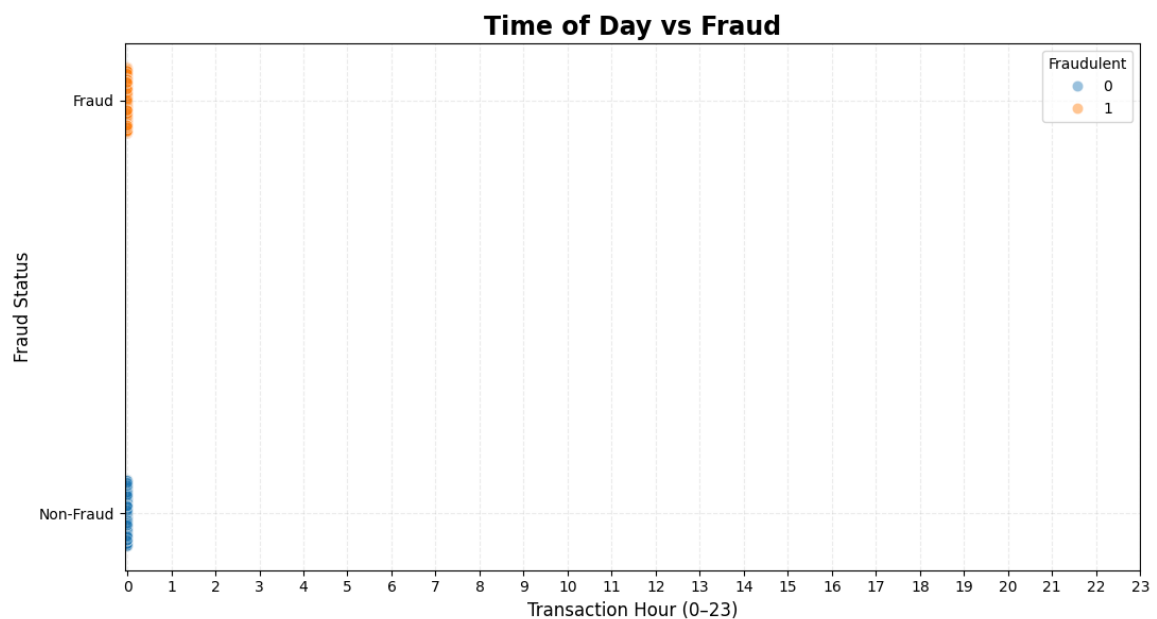
Conclusion

High-value transactions should be treated as a fraud signal, but not as automatic proof of fraud.

11. TIME OF DAY VS FRAUD

Transaction time was converted into hour-based information and analysed against fraud occurrence.

The project uses time-based visualizations to investigate whether fraudulent transactions are concentrated during particular hours.



Conclusion

The analysis provides an exploratory view of time-based fraud behaviour. However, the available notebook results do not establish a statistically confirmed single peak fraud hour.

Therefore, time of day can be used as a supporting monitoring feature rather than a standalone fraud rule.

12. DEVICE CHANGES VS FRAUD

Device behaviour was analysed by examining the number of devices associated with users.

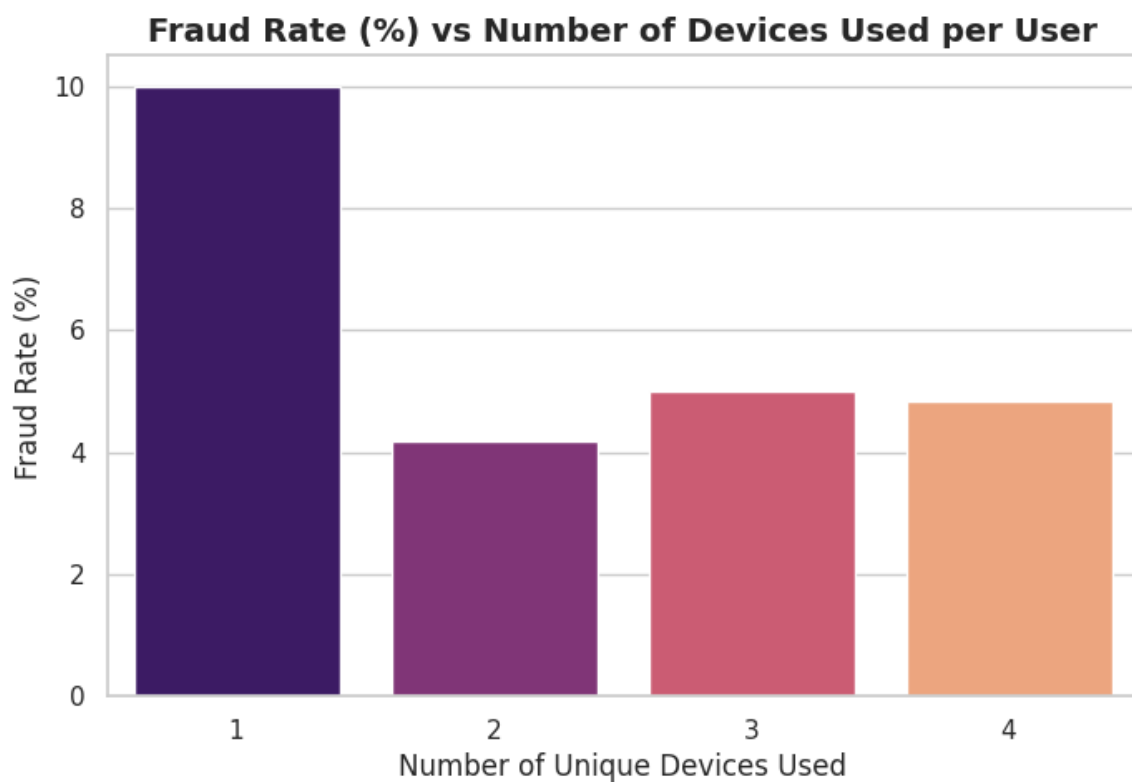
The multiple-device-change signal contains:

- Records: **49,699**
- Fraudulent records: **2,455**
- Fraud rate: approximately **4.94%**

Interpretation

Frequent device changes can indicate unusual user behaviour. However, because this signal covers a very large number of records, it should not be considered sufficient by itself.

Device behaviour can be combined with transaction amount, account age and transaction frequency to improve risk assessment.



13. ACCOUNT AGE VS FRAUD

Account age was analysed to determine whether newer accounts show abnormal behaviour.

A targeted analysis identified:

117 new-account records with abnormal activity.

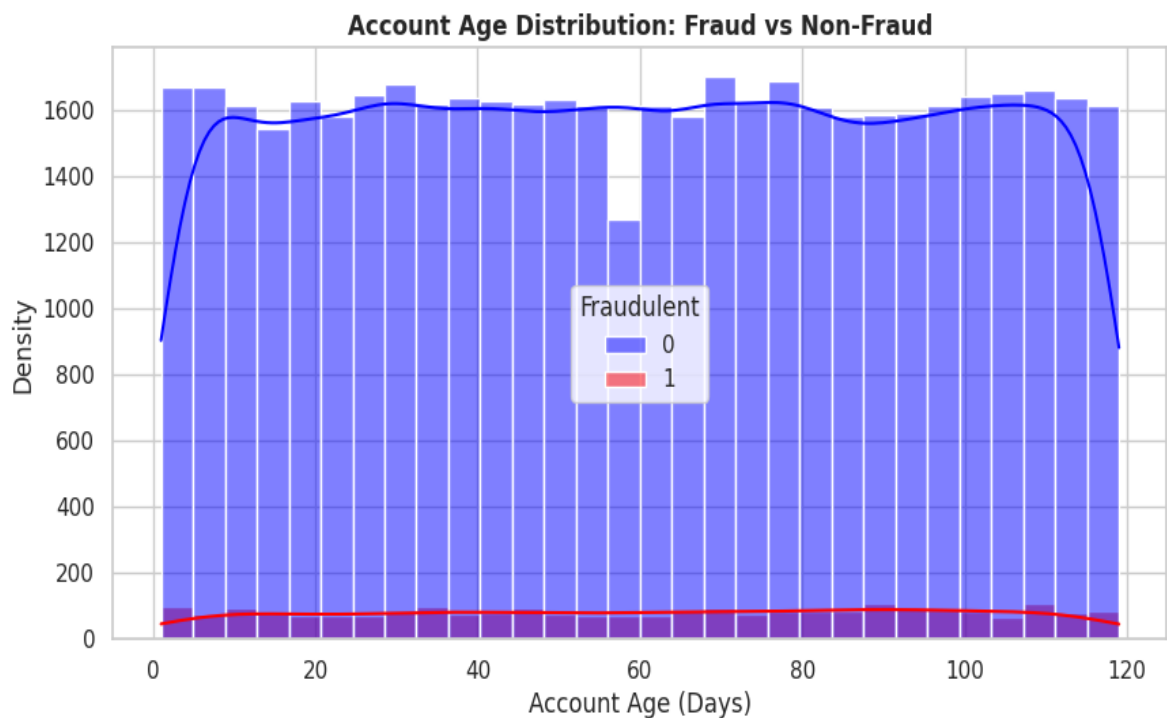
Among these:

7 transactions were fraudulent.

Fraud rate within this group:

5.98%

The overall fraud rate is 4.92%.



Conclusion

The new-account abnormal activity group shows a slightly higher fraud rate than the overall dataset. Therefore, newly created accounts performing unusual transactions should receive additional monitoring.

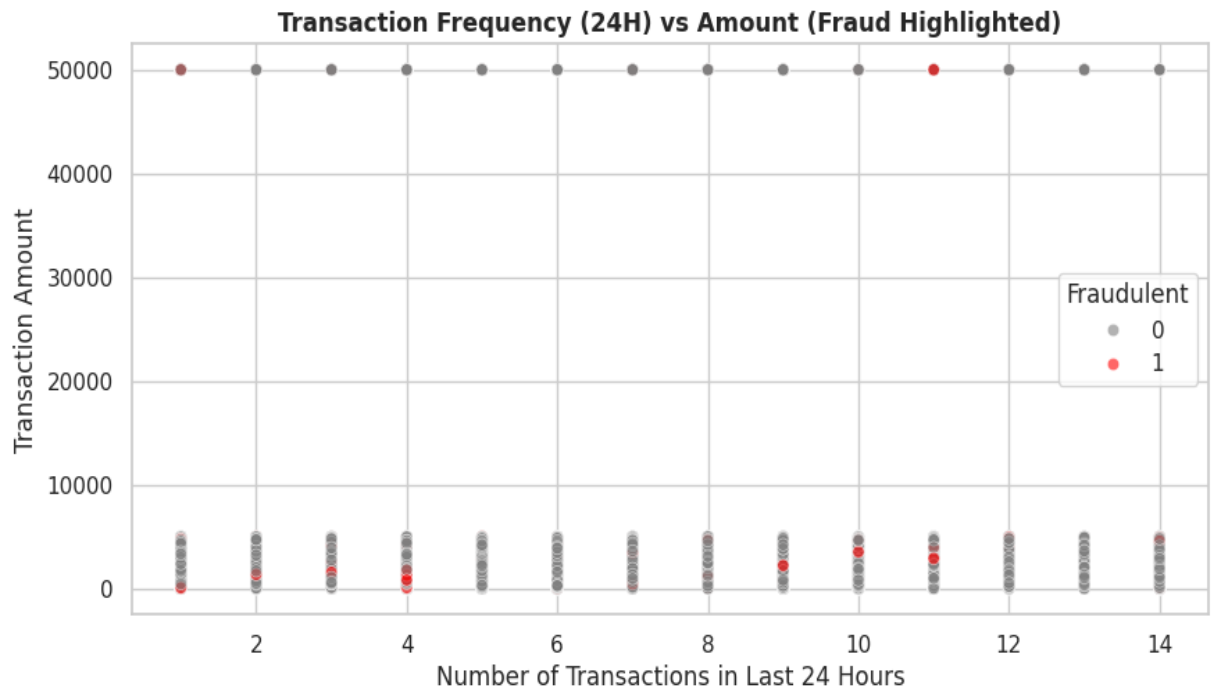
14. TRANSACTION FREQUENCY VS FRAUD

The number of transactions during the previous 24 hours was analysed against transaction amount and fraud status.

Frequent transactions occurring together with unusually high transaction amounts can provide a stronger indication of suspicious behaviour.

Interpretation

Transaction frequency should not be analysed independently. Combining frequency with transaction amount, account age and device behaviour can provide a more useful fraud-risk signal.



15. HYPOTHESIS TESTING

Three statistical tests were performed:

- T-Test
- Chi-square Test
- ANOVA

15.1 T-TEST

Hypotheses

Null Hypothesis (H_0):

Transaction amount has no significant relationship with fraud occurrence.

Alternative Hypothesis (H_1):

Transaction amount has a significant relationship with fraud occurrence.

Result

- T-statistics = **1.0874**
- p-value = **0.27698**

Since:

p-value > 0.05

The null hypothesis is **not rejected**.

Business Interpretation

Transaction amount alone does not provide statistically significant evidence that a transaction is fraudulent.

Therefore, high-value transactions should be combined with other behavioral signals.

OUTPUT

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HYPOTHESIS TEST 1: Two-Sample T-Test (Amount vs Fraud)

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Fraud Mean Amount: \$3118.37

Non-Fraud Mean Amount: \$2989.93

T-Statistic: 1.0874

p-value: 2.7698e-01

Result: Fail to Reject H_0 (No significant difference in average amounts)

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HYPOTHESIS TEST 2: Chi-Square Test (Device Used vs Fraud)

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Chi-Square Statistic: 4.7481

Degrees of Freedom: 3

p-value: 1.9120e-01

Result: Fail to Reject H_0 (Device selection has no relation to fraud)

16. CHI-SQUARE TEST

The Chi-square test was used to examine the relationship between device type and fraud.

Result

- Chi-square = **4.7481**
- Degrees of freedom = **3**
- p-value = **0.19120**

Since the p-value is greater than 0.05, the null hypothesis is not rejected.

Interpretation

Broad device type does not show a statistically significant relationship with fraud in this analysis.

However, detailed device-change behaviour can still be useful as a supporting fraud-monitoring signal.

17. ANOVA TEST

ANOVA was used to compare transaction amounts across payment-method groups.

Result

- F-statistics = **0.3734**
- p-value = **0.82775**

Since:

p-value > 0.05

There is no statistically significant difference in transaction amount across the payment-method groups analysed.

Interpretation

Payment method alone does not provide strong statistical evidence for identifying fraudulent transactions.

18. STATISTICAL FRAUD SIGNALS

The project identifies the following potential fraud signals:

Fraud Signal	Records	Fraud Hits	Fraud Rate
Sudden high-value transfers	508	32	6.30%
New accounts + abnormal activity	117	7	5.98%
Multiple device changes	49,699	2,455	4.94%
Rapid transaction bursts	0	0	0.00%
Late-night transaction spikes	0	0	0.00%

Important Finding

The highest fraud rate among the triggered signals is observed for:

Sudden high-value transfers — 6.30%

This is followed by:

New accounts with abnormal activity — 5.98%

19. BUSINESS INSIGHTS

1. Which variables strongly indicate fraud?

Extreme transaction amounts, abnormal activity in new accounts, device behaviour and transaction frequency are useful indicators when considered together.

2. Are high-value transactions always risky?

No.

The t-test does not show a statistically significant difference between fraud and non-fraud transaction amounts. Therefore, high-value transactions should trigger additional investigation rather than automatic fraud classification.

3. What user behaviours appear suspicious?

The main suspicious behaviours include:

- Extremely high transaction amounts
- Abnormally frequent transactions
- Multiple device usage

- Unusual activity from new accounts
- Combination of high transaction amount and high frequency

4. Which customer segments require monitoring?

The following groups should receive additional monitoring:

- New accounts perform abnormal transactions.
- Users generating extreme transaction amounts.
- Users with unusual transaction frequency
- Users showing repeated device changes.

5. What recommendations should be provided to fraud teams?

Fraud teams should use a **multi-signal risk assessment system** rather than relying on a single variable.

20. RECOMMENDATIONS

1. **Monitor high-value outliers** using IQR and Z-score thresholds.
 2. **Monitor new accounts** that perform unusually large or frequent transactions.
 3. **Combine multiple signals** such as amount, frequency, account age and device behaviour.
 4. **Use device changes as a supporting indicator**, not as the only fraud rule.
 5. **Monitor transaction-time patterns** continuously.
 6. **Create a risk score** based on the severity of multiple suspicious behaviours.
 7. **Reduce false positives** by requiring multiple risk signals before manual investigation.
 8. **Validate fraud rules** using future transaction data before deploying them in a real payment system.
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21. CONCLUSION

The **UPI Fraud Detection** project demonstrates how statistical analysis can be used to identify suspicious transaction behaviour.

The analysis was performed on **50,119 transactions**, of which **2,467 were fraudulent**, representing **4.92%** of the dataset.

The transaction amount distribution is highly skewed, with a skewness of **8.45**. The IQR method identified **508 high-value outliers**, while abnormal activity analysis identified **117 new-account cases**.

Hypothesis testing showed that transaction amount, broad device type and payment method do not individually provide statistically significant evidence of fraud. This demonstrates the importance of using multiple behavioural signals instead of depending on a single feature.

Overall, the project shows that **high-value transactions, abnormal new-account activity, transaction frequency and device behaviour** can be used as fraud-monitoring signals. A combined statistical risk-scoring approach can help fraud teams identify suspicious transactions earlier and focus investigation efforts on the highest-risk cases.

22. FUTURE SCOPE

Future improvements can include:

- Developing a machine-learning fraud prediction model.
- Creating a real-time fraud detection system.
- Building a Power BI fraud monitoring dashboard.
- Developing a combined fraud-risk score.
- Testing the signals on a separate validation dataset.
- Monitoring changes in fraud behaviour over time.
- Improving false-positive detection.
